

PartnerAPI

Project Plan

Software Engineering Project

October 6, 2026

Contents

1	Project Objective	2
2	Work Breakdown Structure	2
3	Estimation-Based Schedule	3
4	Milestones	4
5	Critical Path	4
6	Resource Allocation	5
7	Schedule Constraint	5
8	Assumptions	6
9	Schedule Monitoring	7
10	Conclusion	7

1 Project Objective

The objective of the PartnerAPI project is to prepare the platform for the first five partner agencies within the 10-week target.

The project covers requirements engineering, software design, API development, sandbox preparation, documentation, testing, and partner onboarding.

2 Work Breakdown Structure

1. Requirements Engineering

- (a) Stakeholder identification
- (b) Document analysis
- (c) Stakeholder workshop
- (d) Requirements specification
- (e) MoSCoW prioritisation

2. Software Design

- (a) API architecture
- (b) Use case model
- (c) Class model
- (d) Sequence model
- (e) Activity model
- (f) State model

3. API Development

- (a) Authentication
- (b) Hotel search
- (c) Availability
- (d) Booking
- (e) Cancellation
- (f) Rate limiting
- (g) Usage reporting

4. Sandbox

- (a) Sandbox environment
- (b) Test data
- (c) Sandbox credentials

5. Documentation

- (a) API reference
- (b) Authentication guide
- (c) Error guide
- (d) Partner integration guide

6. Testing

- (a) Functional tests
- (b) Boundary tests
- (c) Decision-table tests
- (d) Security tests
- (e) Integration tests
- (f) Regression tests

7. Partner Onboarding

- (a) Partner credentials
- (b) Sandbox integration
- (c) Testing
- (d) Production activation

3 Estimation-Based Schedule

The project estimates use the supplied three-point PERT estimates. The resulting expected times (TE) for the sequential activities are shown in Table 1.

Table 1: PERT-Based Activity Estimates

Activity	TE (days)
API Design	6.33
Booking Endpoints	15.00
Sandbox	6.67
Documentation	5.17
Partner Onboarding	8.67
Total	41.84

The total sequential estimate is:

41.84 working days

Assuming five working days per week:

$$\frac{41.84}{5} = 8.37 \text{ weeks}$$

The baseline therefore provides approximately 1.63 weeks of schedule margin against the 10-week target.

4 Milestones

Milestone	Completion Condition
Requirements Baseline	SRS approved
Architecture Baseline	UML package approved
API Development Complete	Required endpoints implemented
Sandbox Ready	Partners can test integration
Documentation Complete	API documentation published
Testing Complete	Required tests passed
Partner Readiness	First five partners validated
Production Launch	First five partners activated

5 Critical Path

Under the required sequential estimation assumption, the activities form a single sequential chain:

API Design → Booking Endpoints → Sandbox → Documentation → Partner Onboarding

Therefore, the sequential critical-path duration is:

41.84 working days

or:

8.37 weeks

The baseline estimate therefore remains within the required 10-week delivery target.

6 Resource Allocation

Table 3: Project Resource Allocation

Role	Primary Responsibility
Business Analyst	Requirements
Software Architect	Architecture and UML
Backend Engineer	API implementation
QA Engineer	Testing and quality
DevOps Engineer	Sandbox and deployment
Technical Writer	Documentation
Partner Engineer	Partner onboarding
Project Manager	Schedule, scope and risks

7 Schedule Constraint

The baseline estimate is:

41.84 working days

This corresponds to:

8.37 weeks

when using a five-day working week.

Therefore, the baseline scope fits within the required **10-week target**.

However, the project includes two custom feature requests.

These requests are not part of the common requirements of the partner agencies.

The estimated additional effort is shown in Table 4.

Table 4: Custom Request Impact

Custom Request	Effort	Impact
Custom Request 1	12 days	Additional scope
Custom Request 2	18 days	Additional scope
Combined	30 days	Additional scope

If both custom requests are accepted within the same sequential delivery scope, the estimated project duration becomes:

$$41.84 + 30 = 71.84 \text{ working days}$$

Assuming five working days per week:

$$\frac{71.84}{5} = 14.37 \text{ weeks}$$

The resulting delay against the original 10-week target is:

$$14.37 - 10 = 4.37 \text{ weeks}$$

Therefore, accepting both custom requests without changing the schedule or available resources would create a significant delivery risk.

Possible management responses include:

- Deferring the custom requests to a later release.
- Allocating additional development resources.
- Developing selected features in parallel.
- Obtaining formal approval for a revised deadline.

For the baseline release, the recommended approach is to keep the scope focused on the common requirements of the partner agencies.

The two custom requests should be handled through formal change control or scheduled for a later phase.

8 Assumptions

The schedule is based on the following assumptions:

1. Estimates represent working days.
2. A working week contains five working days.
3. The listed baseline activities are treated as sequential for the purpose of the supplied estimation exercise.
4. The first five partner agencies are the initial production onboarding target.
5. The PERT values are estimates rather than guaranteed delivery dates.
6. Custom requests are treated as additional scope and are not included in the baseline 10-week estimate.

9 Schedule Monitoring

The project should be monitored against the baseline estimate throughout implementation.

Key monitoring areas include:

- Completion of each milestone.
- Actual effort versus estimated effort.
- Partner onboarding progress.
- Testing and defect status.
- Scope changes.
- Security and integration risks.
- Remaining schedule margin.

Any significant deviation should be recorded through the project's change-management process.

Its effect on time, cost, quality, and scope should then be assessed.

10 Conclusion

The baseline PartnerAPI project is estimated to require **41.84 working days**.

This is equivalent to approximately **8.37 weeks** under the supplied sequential estimation assumption.

The baseline therefore fits within the required **10-week target**, leaving approximately **1.63 weeks of schedule margin**.

However, adding both custom requests would increase the estimated sequential effort to **71.84 working days**.

This corresponds to approximately **14.37 weeks**.

The resulting delay would therefore be approximately **4.37 weeks**.

The schedule should be actively monitored, and custom requirements should be managed through formal change control.

The PERT estimate should be treated as a planning estimate rather than a delivery guarantee.